

Complex Dynamics and High codimension Bifurcations of a Predator Prey Model

Wael El Khateeb

University of Toledo

In collaboration with Dr. Chunhua Shan, Dr. Guihong Fan, and Dr. Hao Wang.



History



Adriatic Sea: East of Italy

During World War I, fishing in regions such as the Adriatic Sea declined dramatically, leading to unusual fluctuations in fish populations. Vito Volterra became intrigued by this phenomenon.

Real-Life Data



Classical Model

The Lotka–Volterra model is a pair of first-order, nonlinear differential equations that describe the dynamics of predator and prey populations. Let $x(t)$ represent the prey population and $y(t)$ represent the predator population. The model is given by:

$$\begin{cases} \frac{dx}{dt} = \alpha x - \beta xy, \\ \frac{dy}{dt} = \delta xy - \gamma y, \end{cases} \quad (1)$$

where:

- α is the natural growth rate of the prey,
- β is the predation rate coefficient,
- δ is the growth rate of predators per prey consumed, and
- γ is the natural death rate of the predators.

Extension of Classical Model

Instead of exponential growth, logistic growth is used to account for resource limitations:

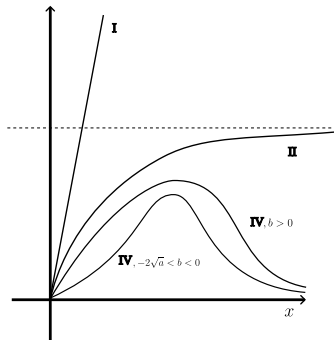
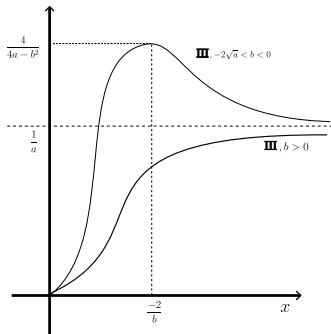
$$\frac{dx}{dt} = \alpha x \left(1 - \frac{x}{K}\right) - p(x)y,$$

where K is the carrying capacity. We can incorporate the natural harvesting of prey by subtracting h :

$$\frac{dx}{dt} = \alpha x(t) \left(1 - \frac{x(t)}{K}\right) - p(x)y - h.$$

Holling Functional Response

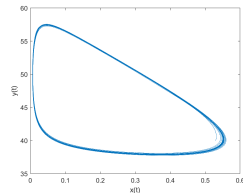
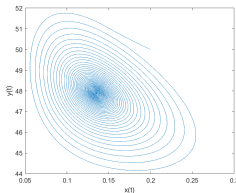
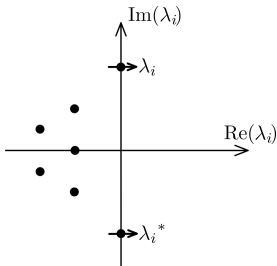
Describe how a predator's rate of prey consumption changes with prey density. They were developed by Canadian ecologist C. S. Holling in the 1950s and 1960s.



Local Hopf Bifurcation

Definition

A *Hopf bifurcation* occurs when a fixed point of a dynamical system loses stability as a pair of complex conjugate eigenvalues crosses the imaginary axis, resulting in the emergence or disappearance of a periodic orbit of small amplitude.



Translation of the predation action

Predators need time to reproduce after predation, which is why we add a delay.

$$\begin{cases} \frac{dx}{dt} = rx(1 - \frac{x}{K}) - p(x(t - \tau))y, \\ \frac{dy}{dt} = p(x(t))y - dy. \end{cases} \quad (2)$$

τ represents the time required for the predator's consumption of prey to translate into an increase in the predator population. The mortality of predators during this period is accounted for by the function $g(\tau, d)$:

$$\begin{cases} \frac{dx}{dt} = rx(1 - \frac{x}{K}) - p(x(t))y, \\ \frac{dy}{dt} = g(\tau, d)p(x(t - \tau))y - dy. \end{cases} \quad (3)$$

Functional Differential Equation

Define

$$C = C([-r, 0], \mathbb{R}^n)$$

Let

$$\sigma \in \mathbb{R}, r, A \in \mathbb{R}^+, x \in C([\sigma - r, \sigma + A], \mathbb{R}^n), \text{ and } t \in [\sigma, \sigma + A]$$

We define $x_t(\theta) = x(t + \theta)$, $\theta \in [-r, 0]$. Assume Ω is a subset of $\mathbb{R} \times C$, $f: \Omega \rightarrow \mathbb{R}^n$ is a given function, then we call:

$$\dot{x}(t) = f(t, x_t) \tag{4}$$

a retarded delay differential equation or functional differential equation. Initial data for the delay differential equations are generally continuous functions defined on a finite interval.

Existence and Uniqueness of Solutions

Theorem (Existence)

Suppose that Ω is an open subset in $\mathbb{R} \times C$ and f is continuous on Ω . If $(\sigma, \phi) \in \Omega$, then there is a solution of (1.1) passing through (σ, ϕ) .

Theorem (Uniqueness)

Suppose Ω is an open set in $\mathbb{R} \times C$, $f: \Omega \rightarrow \mathbb{R}^n$ is continuous and $f(t, \phi)$ is Lipschitz in ϕ in each compact set in Ω . If $(\sigma, \phi) \in \Omega$, then there is a unique solution of Equation 4 through (σ, ϕ)

Note: We say $f(t, \phi)$ is Lipschitz in ϕ in a compact set K of $\mathbb{R} \times C$ if there is a constant $k > 0$ such that for any $(t, \phi_i) \in K, i = 1, 2$,

$$|f(t, \phi_1) - f(t, \phi_2)| \leq k|\phi_1 - \phi_2|$$

Linearization of DDE

$$\begin{cases} \frac{dx}{dt} = f(x, y, x(t-\tau), y(t-\tau)) \\ \frac{dy}{dt} = g(x, y, x(t-\tau), y(t-\tau)) \end{cases} \rightarrow \lambda^2 + C_1\lambda + e^{-\lambda\tau}(C_2\lambda + C_3) + C_4 = 0$$

For Simplicity, we use:

$$D(\lambda, \tau) = P(\lambda, \tau) + e^{-\lambda\tau} Q(\lambda, \tau)$$

Possibly infinitely many eigenvalues satisfy the characteristics equation.

Finding Bifurcation points

We are interested in finding the delay thresholds that get us pure imaginary eigenvalues, ie, $\lambda = i\omega$. We replace λ by $i\omega$ in the characteristics equation:

$$\lambda^2 + C_1\lambda + e^{-\lambda\tau}(C_2\lambda + C_3) + C_4 = 0 \rightarrow -\omega^2 + C_1i\omega + e^{-i\omega\tau}(C_2i\omega + C_3) + C_4 = 0.$$

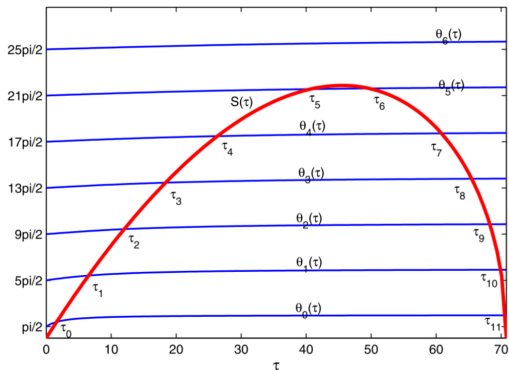
Using Euler's Formula and the trigonometric identity, we get:

$$\begin{cases} \cos(\omega\tau) = -\frac{C_1C_2\omega^2 - C_3\omega^2 + C_3C_4}{C_2^2\omega^2 + C_3^2}, \\ \sin(\omega\tau) = \frac{\omega(C_2\omega^2 + C_1C_3 - C_2C_4)}{C_2^2\omega^2 + C_3^2}, \end{cases} \quad \text{and} \quad \frac{\omega^4 + (C_1^2 - 2C_4)\omega^2 + C_4^2}{C_2^2\omega^2 + C_3^2} = 1.$$

$$\text{Define } S = \omega(\tau)\tau, \text{ and } \theta_n(\tau) = 2n\pi + \arccos\left(-\frac{C_1C_2\omega^2 - C_3\omega^2 + C_3C_4}{C_2^2\omega^2 + C_3^2}\right)$$

Finding Bifurcation points 2

Accordingly, we are interested in finding the points (ω, τ) that satisfy both equations!



Analysis of the graph at hand

To properly analyze the graph above, we should check our eigenvalues at $\tau = 0$.

We therefore have three possibilities that we can conclude from the graph at hand. Denote by $\tau = c_0$ the value over which $S(\tau)$ attains its maximum.

- if $S(c_0) > \theta_0(c_0)$, the graph has $2k$ intersections between $w\tau$ and θ . Thus we expect $2k$ hopf bifurcation thresholds since the difference $(S' - \theta')$ is different than zero, and hence the transversality condition is satisfied.

Remaining Cases

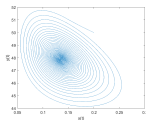
- if $S(c_0) = \theta_0(c_0)$, then there exists one pure imaginary eigenvalue that does not cross to the right side of the plane. Therefore, we do not have Hopf bifurcation.
- If $S(c_0) < \theta_0(c_0)$, then the two graphs of S and θ do not intersect. this means that all eigenvalues would stay on the left side of the plane. $N=N^*$ would be globally asymptotically stable.

Condition for Hopf Bifurcation

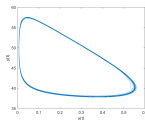
Theorem (Beretta and Kuang(2002))

$$\operatorname{sign}\left\{\left.\frac{d}{d\tau}\operatorname{Re}(\lambda)\right|_{\lambda=i\omega(\tau^*)}\right\}=\operatorname{sign}\left\{\left.\left(\theta'-(\omega(\tau)\tau)'\right)\right|_{\tau=\tau^*}\right\}$$

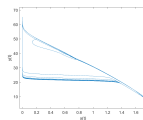
Global Hopf Bifurcation



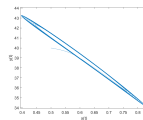
$\tau = 0.5$



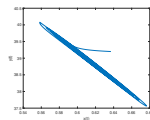
$\tau = 0.7$



$\tau = 8.5$

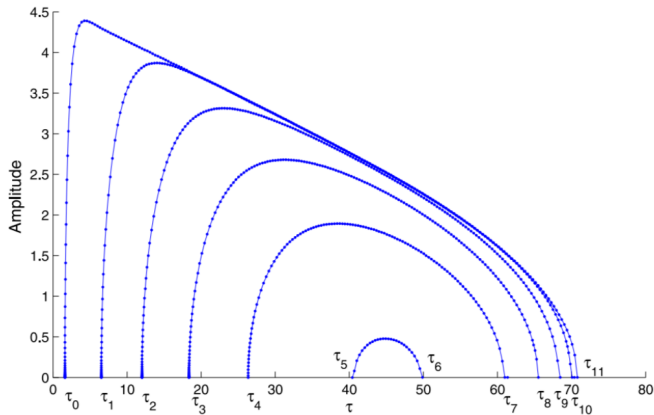


$\tau = 14.4$



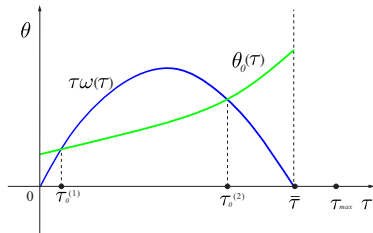
$\tau = 14.8$

Global Hopf Bifurcation 2

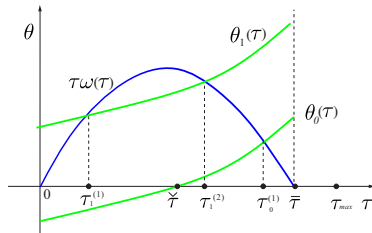


Novel results 1:

What if we have Hopf bifurcation at the classical model level?

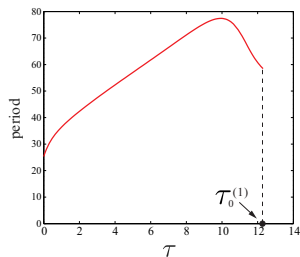
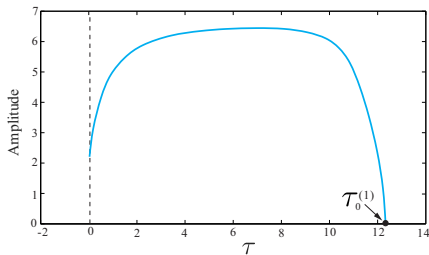


$$K < K_0$$

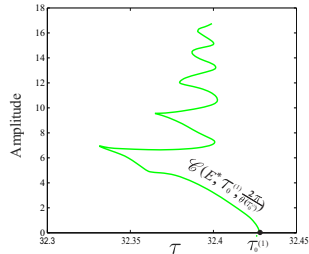
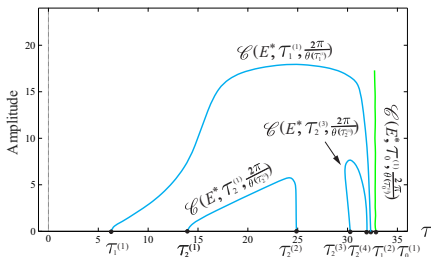


$$K > K_0$$

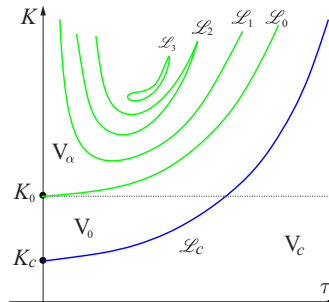
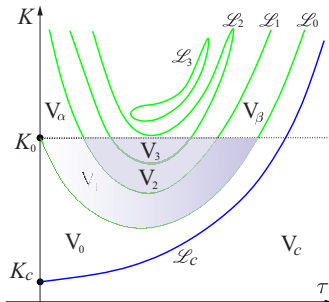
Continuity of branches



Novel Results 2



Novel Results 3



References

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